

Midterm 1 (practice version - NOT GRADED), Math 170a, Fall 2018
Instructor: Elizaveta Rebrova

Printed name: _____

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Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem only: please provide short answers (or short explanations when appropriate).

- (a) (2 points) If events A and B satisfy $\mathbf{P}(A) = 0.5$ and $\mathbf{P}(A \cap B^c) = 0.2$, find the conditional probability $\mathbf{P}(B|A)$.

Solution:

$$\mathbf{P}(B|A) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(A)} = \frac{\mathbf{P}(A) - \mathbf{P}(A \cap B^c)}{\mathbf{P}(A)} = \frac{0.3}{0.5} = 0.6$$

- (b) (2 points) If A is an event with a strictly positive probability what is $\mathbf{P}(A|A)$?

Solution: Use the definition of conditional probability

$$\mathbf{P}(A|A) = \frac{\mathbf{P}(A \cap A)}{\mathbf{P}(A)} = \frac{\mathbf{P}(A)}{\mathbf{P}(A)} = 1.$$

- (c) (2 points) If $\Omega = A \cup B$, $\mathbf{P}(A \cap B^c) = 0.6$, $\mathbf{P}(A^c \cap B) = 0.2$ find the probabilities of A and B .

Solution: The condition $A \cup B = \Omega$ implies that $A^c \cap B = A^c$ and $A \cap B^c = B^c$. So we in fact have $\mathbf{P}(A^c) = 0.2$ and $\mathbf{P}(B^c) = 0.6$, from where $\mathbf{P}(A) = 0.8$ and $\mathbf{P}(B) = 0.4$.

- (d) (2 points) Suppose you are invited to 9 parties and you choose to come to each of them independently with probability 0.3. What is the chance you visit no more than 3 parties?

Solution: $\mathbf{P}(\text{visit no more than 3 parties}) = \mathbf{P}(\text{visit no parties}) + \mathbf{P}(\text{visit 1 party}) + \mathbf{P}(\text{visit 2 parties}) + \mathbf{P}(\text{visit 3 parties}) = 0.7^9 + 9 \cdot 0.3 \cdot 0.7^8 + \binom{9}{2} \cdot 0.3^2 \cdot 0.7^7 + \binom{9}{3} \cdot 0.3^3 \cdot 0.7^6 = 0.7^9 + 9 \cdot 0.3 \cdot 0.7^8 + 36 \cdot 0.3^2 \cdot 0.7^7 + 84 \cdot 0.3^3 \cdot 0.7^6$

- (e) (2 points) Roll a fair die (6 sides, no bias) and look at the number that comes up. Consider the events $A = \{\text{the number is even}\}$, $B = \{\text{the number is less or equal than 4}\}$ and $C = \{\text{the number is less or equal to 5}\}$. Show that events A and B are independent, but A and C are not independent.

Solution: Set the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, then $A = \{2, 4, 6\}$, $B = \{1, 2, 3, 4\}$ and $A \cap B = \{2, 4\}$. Since all the outcomes are equally likely $\mathbf{P}(A) = \frac{1}{2}$, $\mathbf{P}(B) = \frac{2}{3}$ and $\mathbf{P}(A \cap B) = \frac{1}{3}$ and indeed $\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$.

Now $C = \{1, 2, 3, 4, 5\}$ and $A \cap C = \{2, 4\}$, so $\mathbf{P}(C) = \frac{5}{6}$, $\mathbf{P}(A \cap C) = \frac{1}{3}$. Therefore, $\mathbf{P}(A \cap C) \neq \mathbf{P}(A) \cdot \mathbf{P}(C)$.

2. (10 points) There are n different cars labeled with numbers 1 to n . These cars are parked in n consecutive parking spaces from left to right in a uniformly random order. What is the probability that the cars labeled 1 and 2 are parked next to each other (meaning that the car 2 is directly to the left or to the right of the car 1)? Simplify your answer.

Solution: Let A_1 be the event that the car 1 is parked to the left of the car 2, and A_2 the event that the car 1 is parked to the right of the car 2. These events are disjoint and we have to find the probability of their union. Let's find the probability of the first event.

There are $n!$ possibilities to park all the cars without any restrictions. How many of these permutations are those in which the car 1 is to the left of the car 2? Well first we can park the car 1 at any position except at the rightmost position, so there are $n - 1$ ways to park it. Then the position of the car 2 is determined. Then we are left with $n - 2$ cars to park in all other places, which can be done $(n - 2)!$ ways. Therefore, there are in total $(n - 1) \cdot (n - 2)! = (n - 1)!$ permutations of cars in which the car 1 is to the left of the car 2. Therefore,

$$\mathbf{P}(A_1) = \frac{(n - 1)!}{n!} = \frac{1}{n}.$$

In the same way we get $\mathbf{P}(A_2) = \frac{1}{n}$ and so the wanted probability is $2/n$.

3. (10 points) A certain department has 20 employees, 10 are instructors and 10 are TAs. This department participates in a lottery in which the probability of winning the 1st prize is 0.1, winning the 2nd prize is 0.2 and winning the 3rd prize is 0.3 (and winning no prize is 0.4). The 1st prize is a dinner of 15 people, the 2nd prize is a dinner for 10 people and the 3rd prize is a dinner for 5 people. If the department wins a prize, the people going to prize dinner will be chosen uniformly at random out of all 20 employees. What is the probability that exactly one instructor from this department goes to the prize dinner?

Solution: Let E be an event that exactly one instructor from this department goes to the prize dinner. Let Win_1 be an event that department wins 1st prize, Win_2 be an event that department wins 2nd prize, Win_3 be an event that department wins 3rd prize, Win_0 be an event that department wins no prize.

Using Total Probability theorem,

$$\begin{aligned} \mathbf{P}(E) &= \mathbf{P}(E|Win_1)\mathbf{P}(Win_1) + \mathbf{P}(E|Win_2)\mathbf{P}(Win_2) + \\ &\quad \mathbf{P}(E|Win_3)\mathbf{P}(Win_3) + \mathbf{P}(E|Win_0)\mathbf{P}(Win_0) = \\ &\quad \mathbf{P}(E| \text{dinner for 15 people}) \cdot 0.1 + \mathbf{P}(E| \text{dinner for 10 people}) \cdot 0.2 + \\ &\quad \mathbf{P}(E| \text{dinner for 5 people}) \cdot 0.3 + 0 \end{aligned}$$

Note that exactly one instructor at dinner is not possible if a) there is no dinner or b) dinner includes more than 11 people (since there are only 10 non-instructors at the department).

There are $\binom{20}{k}$ ways to choose k people to go to a department dinner. To have exactly one instructor among them, we need to choose an instructor (10 options) and other $k-1$ participants from TAs in $\binom{10}{k-1}$ ways. So,

$$\begin{aligned} \mathbf{P}(E) &= 0 \cdot 0.1 + \frac{10 \cdot \binom{10}{9}}{\binom{20}{10}} \cdot 0.2 + \frac{10 \cdot \binom{10}{4}}{\binom{20}{5}} \cdot 0.3 + 0 = \\ &\quad \frac{20}{\binom{20}{10}} + \frac{630}{\binom{20}{5}} = \frac{10! \cdot 10! \cdot 20 + 5! \cdot 15! \cdot 630}{20!}. \end{aligned}$$

4. (10 points) A burglar just stole your keychain which has n keys, exactly two of which open your apartment door. He will try to open your door using the keys one by one in a random order. Whenever a key does not open your door, he will not attempt to use this key again. What is the probability that the burglar opens your door on the k th attempt?

Solution: We can do this using the multiplication rule. The probability the burglar fails in the first attempt is $\frac{n-2}{n}$. Given the failure in the first try the probability of the failure in the second try is $\frac{n-3}{n-1}$. Given the failure in the first two tries the probability of the failure in the third try is $\frac{n-4}{n-2}$ and so on. Given the failure in the first $k-2$ tries the probability of the failure in the next try is $\frac{n-k}{n-k+2}$. Given the failure in the first $k-1$ tries the probability of the success in the next try is $\frac{2}{n-k+1}$. This gives the wanted probability

$$\frac{2(n-2)(n-3)\cdots(n-k)}{n(n-1)\cdots(n-k+1)} = \frac{2(n-k)}{n(n-1)}.$$

5. (10 points) There are three museums in a town. Each is open with probability 0.4 and closed with probability 0.6 independently of the other museums. Two of the museums are located on the street A and one on the street B. A tourist chooses either street A or street B with equal probability and walks down the whole length of the street he chose. He will enter every **open** museum he encounters on his way. Given that he entered exactly one museum, what is the probability he chose to walk down the street A?

Solution: Use the Bayes' rule here. Let A be the event the tourist chooses to take street A and B he chooses to take street B. Let C be the event he entered exactly one museum.

Given that he takes the street B he will enter exactly one museum if and only if the museum on the street B is open. This has probability 0.6, so $\mathbf{P}(C|B) = 0.4$.

Given that he takes the street A he will enter exactly one museum if and only if exactly one of the two museums on the street A is open. The probability of this is $2 \cdot 0.6 \cdot 0.4 = 0.48$. Therefore, $\mathbf{P}(C|A) = 0.48$.

Since $\mathbf{P}(A) = \mathbf{P}(B) = 0.5$ use the Bayes' rule

$$\mathbf{P}(A|C) = \frac{\mathbf{P}(A)\mathbf{P}(C|A)}{\mathbf{P}(A)\mathbf{P}(C|A) + \mathbf{P}(B)\mathbf{P}(C|B)} = \frac{0.5 \cdot 0.48}{0.5 \cdot 0.48 + 0.5 \cdot 0.4} = \frac{6}{11}.$$